

Total No. of Questions : 8]

[Total No. of Printed Pages : 1

Roll No

AU/IP/IEM/PR-503 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Metrology Measurement And Control
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. Calculate the limits of tolerances and allowances for a 25mm shaft and hole pair designated H_8d_9 .
2. Define the term “Metrology” as applied to engineering industry. Discuss its widening scope after the first world war.
3. Discuss the principal of Interchangeability.
4. Sketch two forms of sine bars in general use. List the possible sources of errors in sine bar and derive an expression of the error in angle setup resulting from an errors in centre distance between the rollers?
5. Describe mechanical comparator and clearly. Explain the magnification method adopted in it.
6. Discuss various types of gauges. How are they designed?
7. Name and describe any three methods used for measuring surface.

OR

Determine the dimension of the maximum chord over four teeth when the gear under inspection has the following specifications.

No. of teeth = 32

Module = 4 mm

Pressure angle = $14\frac{1}{2}^\circ$

8. Write down short notes on any two of the followings:
 - a) Dial gauge
 - b) Tool maker microscope
 - c) CMM
